

Derivation of the Cart-Pole Equations of Motion

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1 Nomenclature

Symbol	Meaning	Unit
x	Cart position, measured in the ground frame	m
θ	Pole angle from upright, counter-clockwise positive	rad
x_p, y_p	Position of the pole's centre of mass	m
M	Mass of the cart	kg
m	Mass of the pole	kg
l	Distance from the pivot to the pole's centre of mass	m
I_S	Moment of inertia of the pole about its centre of mass	kg m ²
g	Gravitational acceleration	m s ⁻²
F	Control force applied to the cart	N
dF	External disturbance force at the pole's centre of mass	N
b	Viscous damping coefficient of the cart	N s m ⁻¹
H, V	Horizontal and vertical reaction forces at the pivot	N
N	Normal force from the ground on the cart	N

The pole is modelled as a uniform slender rod. With l defined as *half* the rod length, its moment of inertia about the centre of mass is

$$I_S = \frac{1}{12} m (2l)^2 = \frac{1}{3} m l^2. \quad (1)$$

2 The cart-pole problem: definitions

Considering the cart-pole problem as depicted in Figure 1, the first step is to find the position of the pole's centre of mass as a function of the tilt θ and the cart position x . Because the origin is fixed in the ground rather than attached to the cart, the cart's own position must appear in the expression:

$$x_p = x - l \sin \theta, \quad (2)$$

$$y_p = l \cos \theta. \quad (3)$$

Newton's second law relates forces to *accelerations*, so (2) and (3) must be differentiated twice with respect to time. Since θ is itself a function of time, both the chain rule and the

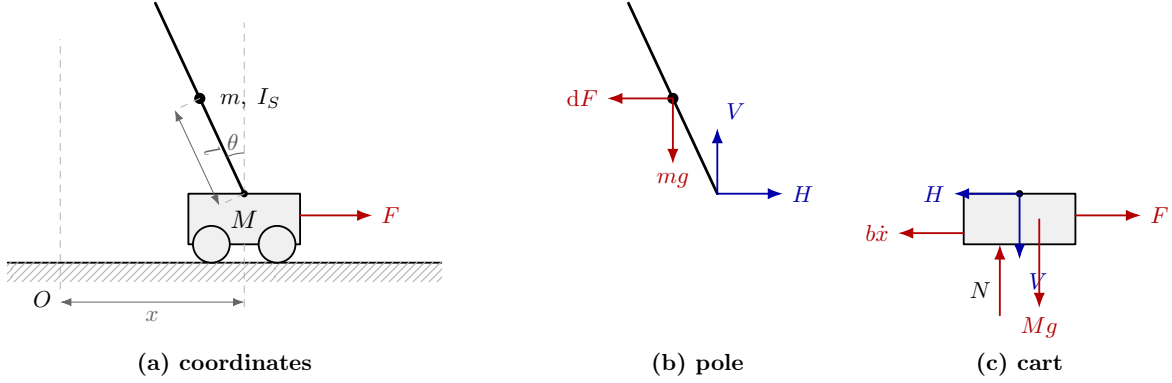


Figure 1: Cart-pole system. (a) Coordinates and conventions, (b) free-body diagram of the pole, (c) free-body diagram of the cart.

product rule apply:

$$\dot{x}_p = \dot{x} - l\dot{\theta} \cos \theta, \quad \ddot{x}_p = \ddot{x} - l\ddot{\theta} \cos \theta + l\dot{\theta}^2 \sin \theta, \quad (4)$$

$$\dot{y}_p = -l\dot{\theta} \sin \theta, \quad \ddot{y}_p = -l\ddot{\theta} \sin \theta - l\dot{\theta}^2 \cos \theta. \quad (5)$$

The terms proportional to $\dot{\theta}^2$, together with the trigonometric functions of θ , are what make the system nonlinear.

3 The pole

The system is split into two bodies, the pole (Figure 1b) and the cart (Figure 1c), connected at the pivot. The interaction between them is represented by the reaction forces H and V , which appear on both free-body diagrams with opposite signs.

Applying the Newton–Euler equations to the pole, first the balance of linear momentum (*Schwerpunktsatz*) in both directions,

$$\sum F_x = m \ddot{x}_p \implies H - dF = m \ddot{x}_p, \quad (6)$$

$$\sum F_y = m \ddot{y}_p \implies V - mg = m \ddot{y}_p. \quad (7)$$

Next the balance of angular momentum about the centre of mass (*Drallsatz*). Gravity acts at the centre of mass and therefore produces no moment about it, which is precisely why this reference point is chosen:

$$\sum M_S = I_S \ddot{\theta} \implies I_S \ddot{\theta} = l \sin \theta V + l \cos \theta H. \quad (8)$$

The lever arms follow from elementary trigonometry, as shown in Figure 2. The vector from the centre of mass S to the pivot P is $(l \sin \theta, -l \cos \theta)$. The horizontal force H therefore acts with the *vertical* lever arm $l \cos \theta$, and the vertical force V with the *horizontal* lever arm $l \sin \theta$. With θ counted counter-clockwise, both contributions carry a positive sign.

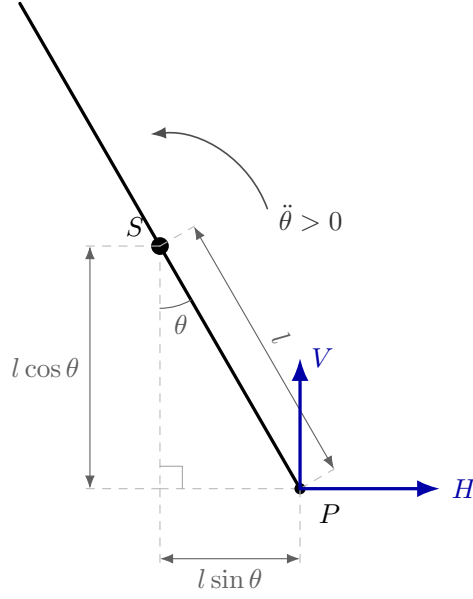


Figure 2: Lever arms of the pivot reactions about the pole's centre of mass.

Solving (6) and (7) for the reaction forces and inserting (4) and (5) gives

$$H = dF + m\ddot{x} - ml\ddot{\theta} \cos \theta + ml\dot{\theta}^2 \sin \theta, \quad (9)$$

$$V = mg - ml\ddot{\theta} \sin \theta - ml\dot{\theta}^2 \cos \theta. \quad (10)$$

Substituting (9) and (10) into the moment equation (8) yields

$$\begin{aligned} I_S \ddot{\theta} &= mgl \sin \theta - ml^2 \ddot{\theta} \sin^2 \theta - ml^2 \dot{\theta}^2 \sin \theta \cos \theta \\ &\quad + dF l \cos \theta + ml\ddot{x} \cos \theta - ml^2 \ddot{\theta} \cos^2 \theta + ml^2 \dot{\theta}^2 \sin \theta \cos \theta. \end{aligned}$$

The two $\dot{\theta}^2$ terms cancel exactly, and the $\ddot{\theta}$ terms combine through $\sin^2 \theta + \cos^2 \theta = 1$. Collecting them on the left-hand side gives the first equation of motion:

$$\boxed{(I_S + ml^2) \ddot{\theta} = ml(g \sin \theta + \ddot{x} \cos \theta) + dF l \cos \theta} \quad (11)$$

The factor $I_S + ml^2$ is the moment of inertia of the pole about the *pivot* rather than about its center of mass, in accordance with Steiner's theorem. It is the point the pole physically rotates around, so it is the quantity that must appear. With (1),

$$I_S + ml^2 = \frac{1}{3}ml^2 + ml^2 = \frac{4}{3}ml^2. \quad (12)$$

4 The cart

The cart translates without rotating, so its angular acceleration vanishes and the moment balance carries no information about the motion. The vertical force balance likewise only determines the normal force N exerted by the ground. Neither is required.

This follows from a degree-of-freedom argument: the system has exactly two independent coordinates, x and θ , and therefore exactly two equations of motion. Every further balance equation determines a constraint force rather than a motion.

The remaining horizontal balance is

$$M\ddot{x} = F - H - b\dot{x}. \quad (13)$$

Substituting H from (9) and collecting the acceleration terms on the left gives the second equation of motion:

$$\boxed{(M + m)\ddot{x} - ml \cos \theta \ddot{\theta} = F - b\dot{x} - dF - ml\dot{\theta}^2 \sin \theta} \quad (14)$$

5 Mass matrix form

Equations (11) and (14) are coupled: both $\ddot{x}$ and $\ddot{\theta}$ appear in each. Sorting all terms containing an acceleration to the left-hand side allows the pair to be written as a linear system,

$$\underbrace{\begin{bmatrix} M + m & -ml \cos \theta \\ -ml \cos \theta & \frac{4}{3}ml^2 \end{bmatrix}}_{\mathbf{A}(\theta)} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F - b\dot{x} - dF - ml\dot{\theta}^2 \sin \theta \\ mgl \sin \theta + dF l \cos \theta \end{bmatrix}. \quad (15)$$

The matrix $\mathbf{A}(\theta)$ is the mass matrix of the system. It plays the same role that the scalar mass plays in $F = ma$, but it is a matrix because the two coordinates are coupled, and it depends on the configuration θ rather than being constant.

The off-diagonal entries $-ml \cos \theta$ quantify that coupling. They vanish at $\theta = \pm\pi/2$: with the pole horizontal, accelerating the cart exerts no moment about the pivot, and the two motions momentarily decouple.

That both off-diagonal entries are identical is not a coincidence. Mass matrices are symmetric for any mechanical system, which makes the symmetry of (15) an independent confirmation that the derivation is consistent.

Inverting the 2×2 matrix gives $\ddot{x}$ and $\ddot{\theta}$ explicitly, which is the form required for numerical integration of the state $\mathbf{x} = [x, \dot{x}, \theta, \dot{\theta}]^T$.